Predicting, Classifying, and Clustering Beer Quality: A Machine Learning Study of Online Beer Reviews

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Abstract

This study investigates whether machine learning techniques can accurately predict beer quality using the Beer Reviews dataset from BeerAdvocate, obtained from OpenML [OpenML, 2026]. Three machine learning tasks were performed: regression, classification, and clustering. Linear Regression, and Random Forest Regression [Breiman, 2001] were compared for predicting overall beer ratings. Logistic Regression, Linear Discriminant Analysis, Quadratic Discriminant Analysis , and Random Forest were evaluated for beer quality classification. K-means clustering was used to identify natural groupings of beers based on sensory attributes. Model performance was evaluated using RMSE, R^2, accuracy, precision, recall, and F1-score. Results indicate that Linear Regression achieved the strongest predictive performance, with an **RMSE of 0.412** and **R^2 of 0.665** for regression, while **Logistic Regression** achieved the highest classification performance with a **macro F1-score of 0.670**. K-means clustering identified **three** interpretable groups of beers representing lower-quality, moderate-quality, and premium higher-ABV beers. Overall, the findings demonstrate that sensory review attributes are strong predictors of beer quality and that machine learning techniques can effectively uncover meaningful patterns within beer review data.

Contents

1 Introduction **3**

2 Data and Exploratory Data Analysis **3**
2.1 Dataset Description . . . 3
2.2 Exploratory Data Analysis . . . 4
2.2.1 Target Variable Distributions . . . 4
2.2.2 Correlation Structure and Data Leakage . . . 4

3 Methods **5**
3.1 Overall Workflow . . . 5
3.2 Software Environment . . . 5
3.3 Task 1: Regression — Predicting Overall Rating . . . 6
3.4 Task 2: Classification — Predicting Rating Tier . . . 6
3.5 Task 3: Unsupervised Learning . . . 6
3.5.1 Principal Component Analysis . . . 7
3.5.2 K-Means Clustering . . . 7
3.6 Model Selection and Validation . . . 7

4 Results **8**
4.1 Task 1: Regression Results . . . 8

4.2 Task 2: Classification Results 9
4.3 Task 3: Unsupervised Learning Results 10
4.3.1 Principal Component Analysis 10
4.3.2 K-Means Clustering 11

5 Conclusion **12**

1 Introduction

Online review platforms have made it possible to study consumer taste at a scale that was previously unimaginable: BeerAdvocate alone hosts millions of consumer-submitted beer reviews, each rating a beer on several sensory dimensions in addition to an overall score [McAuley et al., 2012]. This scale raises a natural empirical question: how well can a beer's overall rating, category, or place among its peers be inferred purely from how it was reviewed? Answering this is of practical interest to breweries seeking to understand what drives consumer appreciation, and to recommendation systems that must summarize a beer's quality from sparse or noisy review signals.

This project investigates that question using approximately 1.58 million BeerAdvocate reviews from a wide range of breweries and beer styles. To improve computational efficiency, a random sample of 50,000 observations was selected for analysis. Exploratory Data Analysis (EDA) was first conducted to examine the distribution of variables, identify potential data quality issues, and understand relationships between the review attributes.

This study investigates two supervised machine learning tasks. Task 1 is a *regression* problem: predicting a beer's overall rating from its sensory sub-scores and alcohol by volume (ABV). Task 2 is a *classification* problem: categorizes beers into pre-defitned quality catergories based on the same predictor variables. Several machine learning models, including Linear Regression, Logistic Regression, K-Nearest Neighbours (KNN), Decision Tree, Naïve Bayes, and Random Forest are implemented and compared to evaluate their predictive performance. The performance of each model is assessed using appropriate evaluation metrics. Regression models are evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2), while classification models are compared using Accuracy, Precision, Recall, F1-score, and confusion matrices. The findings provide insights into the effectiveness of different machine learning algorithms for predicting beer quality and identifying the factors that most strongly influence consumer ratings. The remainder of this paper is organized as follows. Section 2 describes the dataset and presents exploratory analysis of its structure, correlations, and temporal patterns. Section 3 outlines the methodology and machine learning models. Section 4 presents the experimental results and model comparisons, while Section 5 concludes with the main findings and a short discussion of limitations.

2 Data and Exploratory Data Analysis

2.1 Dataset Description

This study uses the BeerAdvocate reviews dataset originally introduced by McAuley et al. [2012] and obtained via the OpenML repository[OpenML, 2026], comprising approximately 1.58 million reviews. Each observation represents a single user review and indicates the variables `review_overall`, `review_aroma`,
`review_appearance`, `review_palate`, and `review_taste` (all measured on a continuous scale from 1 to 5), along with `beer_abv`, `beer_style` (104 categories), `brewery_id`/`brewery_name`, and `review_time`.

Prior to model development, the data set was preprocessed by handling missing values in `the beer _abv` variable, removing duplicate reviews, and examining outlier ABV values. A random sample of 50,000 observations was used for this study to improve computational efficiency while maintaining a representative subset of the original dataset.

The overall rating is used as the target variable for both regression and classification analyzes, while the remaining attributes serve as predictor variables. Exploratory Data Analysis (EDA) was performed to examine the distribution of variables, identify data quality issues, and summarize the characteristics of the dataset. Table 1 presents the descriptive statistics of the numerical variables.

The dataset was then divided into 80% training data and 20% testing data for model training and performance evaluation.

Table 1: Summary statistics for numeric variables.

Variable	Mean	Median	Std. Dev.	Min	Max
review_overall	3.82	4.00	0.72	0.00	5.00
review_aroma	3.74	4.00	0.70	1.00	5.00
review_appearance	3.84	4.00	0.62	0.00	5.00
review_palate	3.74	4.00	0.68	1.00	5.00
review_taste	3.79	4.00	0.73	1.00	5.00
beer_abv	7.04	6.50	2.32	0.01	57.7

2.2 Exploratory Data Analysis

2.2.1 Target Variable Distributions

Figure 1 presents the distribution of the target variable, overall beer rating (`review_overall`). The histogram indicates that the majority of the ratings are concentrated between 3.5 and 4.5, with the highest frequency occuring at approximately 4.0. In contrast, relatively few beers receive ratings below 2.5, indicating that poorly rated beers are uncommon within the dataset.

The distribution is negatively (left) skewed, with observations concentrated at the upper end of the rating scale and a longer tail extending towards lower ratings. This suggests that users generally assign favourable ratings to beers. Consequently, lower-rated beers are underrepresented relative to highly rated beers, which would lead to class imbalance if a binary classification threshold (e.g. 4.0) were adopted. To address this issue, the classification task described in Section 3.4 uses percentile-based rating tiers (Low, Medium, and High) to produce a more balanced target variable.

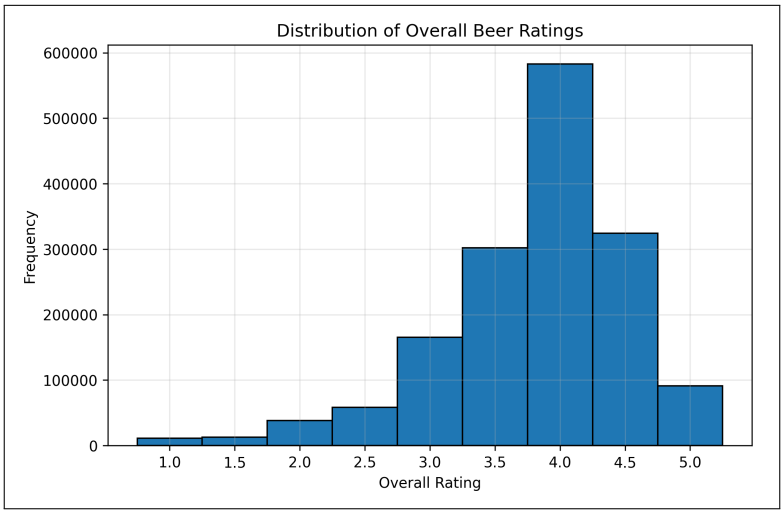


Figure 1: Distribution of overall beer ratings.

2.2.2 Correlation Structure and Data Leakage

Figure 2 represents correlation matrix for the numerical variables in the Beer Reviews dataset. It shows that the sensory review attributes are positively correlated with the overall rating, with

`review_taste` exhibiting the strongest correlation (r = 0.79), followed by `review_palate` (r = 0.70), `review_aroma`, and `review_appearance` (r = 0.50). In contrast, `beer_ABV` shows only a weak positive correlation with the overall rating (r = 0.13), suggesting that alcohol content alone has limited influence on users' overall evaluations. Since the overall rating is assigned after evaluating these sensory characteristics, it is likely to be partially influenced by the individual sub-scores. Consequently, using all four sensory attributes as predictor variables introduces a degree of **target leakage**, meaning that the regression results should be interpreted as an upper bound of predictive performance rather than a measure of how well independent beer characteristics alone predict overall quality. This limitation is acknowledged when interpreting the regression results.

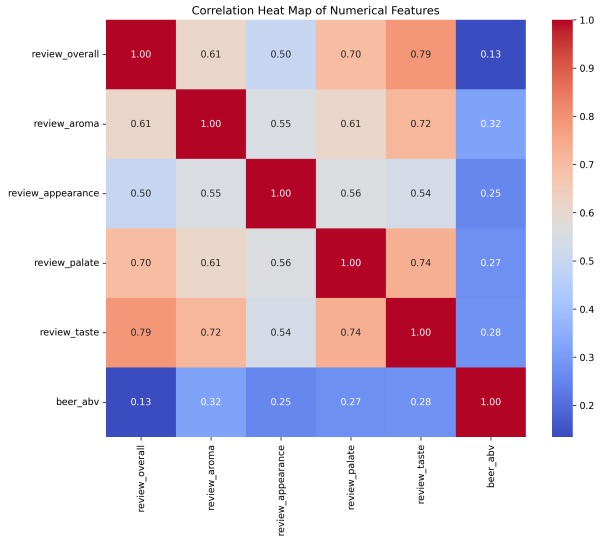


Figure 2: Correlation heatmap of the numerical variables.

3 Methods

3.1 Overall Workflow

This study follows a standard machine learning workflow comprising data pre-processing, model development, and model evaluation. The dataset was first cleaned and explored using exploratory data analysis (EDA) before being divided into training (80%) and testing (20%) sets. Regression, Classification, and Unsupervised Learning models were then developed using training data, with hyperparameter tuning performed through cross-validation where appropriate. Finally, the models were evaluated on the held-out test set using suitable performance metrics, while clustering results were assessed using internal validation measures and visualization techniques.

3.2 Software Environment

All analyses were conducted using Python within the Google Colab environment. Data manipulation and pre-processing were performed using the **Pandas** and **NumPy** libraries, while **Matplolib** and **Seaborn** were used to produce the visualizations presented in the exploratory data analysis. **Scikit-learn**[Pedregosa et al., 2011] library was also used for data pre-processing, model training, cross-validation, hyperparameter tuning and model evaluation. Prior to model development, the dataset was inspected for missing values, duplicate observations, and outliers to ensure the quality of the data was used in the analysis.

3.3 Task 1: Regression — Predicting Overall Rating

The objective of this task was to predict the overall beer rating (`review_aroma`, `review_appearance`, `review_palate`, `review_taste`, and `beer_abv`). As discussed in Section 2.2, the review sub-scores are closely related to the overall rating and my introduce information leakage. Therefore, the regression results should be interpreted as an upper bound on the predictive performance achievable using review-derived numerical features.

An 80:20 train-test split was applied, and missing values in `beer_abv` were imputed using the median. Predictor variables were standardized for Linear Regression, Ridge Regression, and Lasso Regression, while Random Forest Regression was trained using the original feature values, as tree-based models do not require feature scaling. Multiple Linear Regression was used as the baseline model, while Ridge and Lasso Regression were included to examine the effect of L2 and L1 regularization, respectively. Random Forest Regression serves as a nonlinear comparison model.

Hyperparameters for Ridge, Lasso, and Random Forest were tuned using five-fold cross-validation with GridSearchCV. Model performance was evaluated on the held-out test set using Root **Mean Squared Error (RMSE)**, **Mean Absolute Error (MAE)**, and the **coefficient of determination (R^2)**. RMSE was selected as the primary evaluation metric because it penalizes larger prediction errors more heavily than MAE, while 95& bootstrap confidence intervals were calculated to quantify the uncertainty of the RMSE estimates.

3.4 Task 2: Classification — Predicting Rating Tier

As motivated by the target variable distribution observed in Section 2.2, the overall ratings were grouped into Low, Medium, and High rating tiers using the 25th and 75th percentiles of the `review_overall` distribution. This approach produces a more balanced classification problem than a binary threshold based on a fixed rating.

The dataset was divided into an 80:20 train-test split. Missing values in `beer_abv` were imputed using the median, and predictor variables were standardized for Logistic Regression and Linear Discriminant Analysis (LDA). Three classification models were evaluated: Logistic Regression, LDA, and Random Forest Classifier. Logistic Regression minimizes the cross-entropy (log-loss) objective, while Random Forest was included as a nonlinear comparison model. Hyperparameters were selected using five-fold cross-validation with GridSearchCV.

Model performance was evaluated on the held-out test set using accuracy, precision, recall, and macro F1-score. Although accuracy provides an overall measure of classification performance, it may be misleading when class distributions are unequal. Therefore, precision, recall, and macro F1-score were also reported to provide a more balanced assessment across all rating tiers. In addition, a confusion matrix and Receiver Operating Characteristic (ROC) curves were produced for the best-performing model to further evaluate its classification performance.

3.5 Task 3: Unsupervised Learning

The objective of this task was to identify natural groupings of beers based on their review characteristics without using predefined class labels. Clustering was performed using the numerical features `review_aroma`, `review_appearance`, `review_palate`, `review_taste`, and `beer_abv`. The target variable, `review_overall`, was deliberately excluded because it is highly correlated with the review sub-scores (Section 2.2), and including it would bias the clustering towards the overall rating rather than discovering inherent patterns in the feature space.

Missing values in `beer_abv` were imputed using the median, and all features were standardized using `StandardScaler` to ensure equal contribution to the distance calculations used by the clustering algorithms. K-Means clustering was applied as the primary clustering method, with the

optimal number of clusters determined using the elbow method and silhouette analysis. Principal Component Analysis (PCA) was subsequently used to project the standardized feature space into two dimensions for visualization and interpretation of the resulting clusters.

3.5.1 Principal Component Analysis

Principal Component Analysis (PCA) was applied to the standardized sensory features (`review_aroma`, `review_appearance`, `review_palate`, `review_taste`, and `beer_abv`) to reduce the dimensionality of the dataset while preserving as much variation as possible. The proportion of variance explained by each principal component was examined using a scree plot and cumulative explained variance to determine the number of components to retain. The retained components were interpreted using their feature loadings, and the first two principal components were used to produce a two-dimensional visualization of the beers. PCA was used solely for dimensionality reduction and visualization and was not treated as a clustering method.

3.5.2 K-Means Clustering

K-Means clustering[MacQueen, 1967] was applied to the standardized sensory features to identify groups of beers with similar review characteristics. The optimal number of clusters (k) was determined by examining both the elbow method[Rousseeuw, 1987], based on within-cluster sum of squares (inertia), and the silhouette score across a range of candidate k values. Once the optimal number of clusters was selected, the K-Means algorithm was fitted to the standardized dataset.

The resulting clusters were evaluated by comparing their sizes, mean feature values, and overall silhouette score. Cluster profiles were summarized using the average sensory characteristics of each cluster to facilitate interpretation. The clusters were also visualized using the two-dimensional PCA projection to assess their separation and to examine whether they corresponded to existing beer style categories or revealed alternative patterns within the data.

3.6 Model Selection and Validation

Model selection was performed using five-fold cross-validation on the training set, with GridSearchCV used to identify the optimal hyperparameters for the supervised learning models. All models were implemented using scikit-learn[Pedregosa et al., 2011]. The held-out test set was reserved exclusively for final model evaluation and was not used during model training or hyperparameter tuning, ensuring an unbiased estimate of predictive performance.

Overfitting was monitored by comparing model performance on the training and test datasets. Large differences between training and test metrics would indicate poor generalization, whereas similar performance across both datasets suggests that the models generalize well to unseen data. All machine learning models and evaluation procedures were implemented using the `scikit-learn` library.

Table 2: Summary of Tuned Hyperparameters for Best-Performing Models

Task	Hyperparameter	Value
Task 1: Regression (Linear Regression)	Fit Intercept	True
Task 1: Regression (Linear Regression)	Normalization	False
Task 2: Classification (Logistic Regression)	Penalty	l2
Task 2: Classification (Logistic Regression)	C (Regularization Strength)	1.0
Task 2: Classification (Logistic Regression)	Solver	lbfgs
Task 2: Classification (Logistic Regression)	Max Iterations	1000
Task 3: Clustering (K-Means)	k (Number of Clusters)	3
Task 3: Clustering (K-Means)	Initialization	k-means++
Task 3: Clustering (K-Means)	Random State	42

4 Results

4.1 Task 1: Regression Results

Table 3 summarizes the predictive performance of the four regression models on the held-out test set. Overall, all models achieved similar performance, with Linear Regression, Ridge Regression, and Lasso Regression producing the lowest prediction errors (**RMSE = 0.410, MAE = 0.308**) and the highest coefficient of determination (**R^2 = 0.665**). Random Forest Regression achieved a comparable performance (RMSE = 0.412, MAE = 0.311, R^2 = 0.662). Although Random Forest is capable of modelling more complex nonlinear relationships, the linear regression models achieved slightly better predictive performance. This suggests that the relationship between the selected sensory attributes and overall beer rating is predominantly linear, and that the additional flexibility of Random Forest did not provide a meaningful advantage for this dataset.

The 95% confidence intervals for RMSE were narrow and largely overlapped across all models, indicating that the differences in predictive performance were small and unlikely to be practically significant. Consequently, Multiple Linear Regression was selected as the best-performing model because it achieved the lowest prediction error while remaining the simplest and most interpretable approach.

Table 3: Regression Model Performance

Model	RMSE (95% CI)	MAE	R^2
Linear Reg.	0.410 (0.402, 0.419)	0.308	0.665
Ridge Reg.	0.410 (0.402, 0.419)	0.308	0.665
Lasso Reg.	0.410 (0.402, 0.419)	0.308	0.665
Random Forest	0.412 (0.403, 0.420)	0.311	0.662

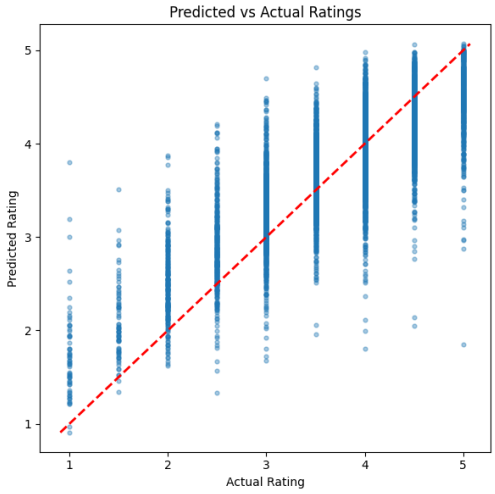


Figure 3: Predicted vs Actual Ratings

Explainability

Figure 3 shows the relationship between the observed and predicted overall ratings. Most observations lie close to the diagonal reference line, indicating that the model predicts the overall ratings with reasonable accuracy. A slight increase in the spread of residuals is visible at the extreme rating values, suggesting that predictions for exceptionally low or exceptionally high ratings are less accurate than those near the center of the distributions. However, no strong systematic bias is evident.

4.2 Task 2: Classification Results

Table 4 compares the performance of the four classification models. Logistic Regression achieved the highest overall performance, with an accuracy of 0.669 (95% CI: 0.660–0.678) and a macro F1-score of 0.670. LDA produced comparable results, whereas Random Forest and QDA performed slightly worse across all evaluation metrics.

The superior performance of Logistic Regression suggests that the relationship between the sensory review attributes and beer rating tiers is largely linear. Although Random Forest can model more complex nonlinear decision boundaries, it did not provide a meaningful improvement over the simpler linear classifier. The confusion matrix (Figure 4) shows that most classification errors occurred between adjacent rating tiers, indicating that beers with similar sensory characteristics are more difficult to distinguish than those belonging to clearly different quality categories.

The ROC and precision–recall curves for the best-performing model are presented in Figure5. Overall, the results indicate that Logistic Regression provides reliable and consistent classification performance despite the class imbalance identified during the exploratory data analysis.

Table 4: Classification Performance Comparison with 95% CI

Model	Accuracy (95% CI)	Precision	Recall	F1-Score
Logistic Regression	0.669 (0.660, 0.678)	0.675	0.669	0.670
Random Forest	0.628 (0.619, 0.637)	0.631	0.628	0.629
LDA	0.664 (0.655, 0.673)	0.697	0.664	0.664
QDA	0.649 (0.639, 0.658)	0.674	0.649	0.653

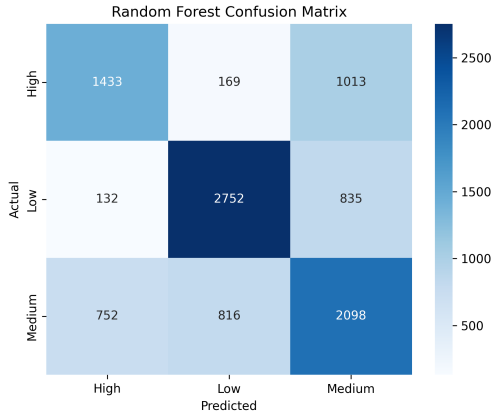


Figure 4: Confusion Matrix

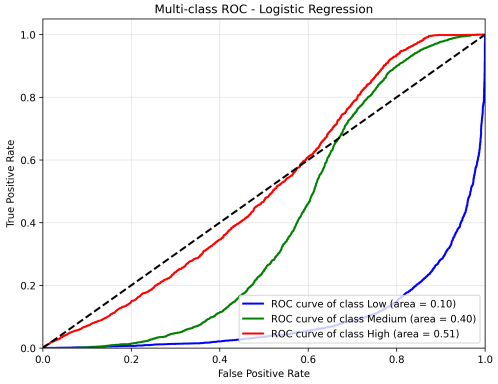


Figure 5: ROC Curve

Explainability

Figure 6 presents the feature importance for the best-performing classification model. Review taste and review aroma contributed most to predicting beer rating tiers, followed by review palate and review appearance, whereas beer ABV had the lowest importance. The similarity between the regression and classification results indicates that flavour-related characteristics consistently have the greatest influence on users' evaluations of beer quality.

Overall, the feature importance pattern closely mirrors the findings from the regression analysis, suggesting that flavour-related sensory characteristics are the primary factors influencing both the predicted overall rating and the classification of beers into rating tiers. This consistency provides additional evidence that taste and aroma are the most influential attributes in users' overall beer evaluations.

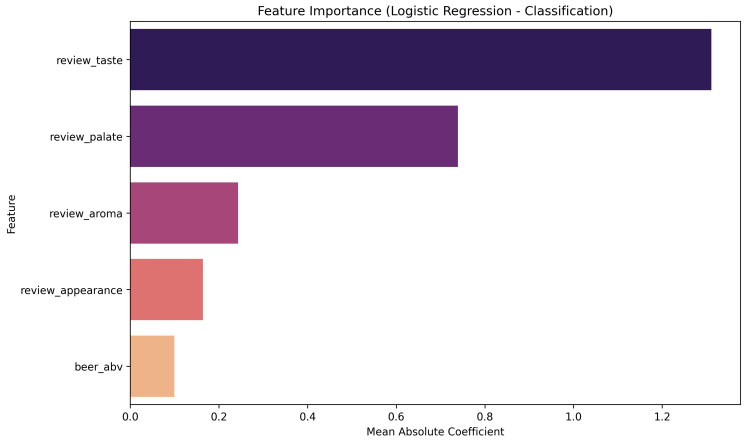


Figure 6: Feature importance for the best-performing classification model (Logistic Regression).

4.3 Task 3: Unsupervised Learning Results

4.3.1 Principal Component Analysis

Principal Component Analysis (PCA) was applied to the standardized sensory variables to reduce dimensionality while preserving the underlying variation in the data. As shown in Table 8a, the first three principal components explained **87.68%** of the total variance, exceeding the commonly used 80% threshold. The first principal component (PC1) accounted for 60.44% of the variance and was dominated by the sensory review attributes (aroma, appearance, palate, and taste), indicating that it represents an overall sensory quality dimension. The second principal component (PC2) explained 16.92% of the variance and was primarily associated with beer ABV, suggesting that alcohol content captures a distinct source of variation. The component loadings are presented in Table 8b.

Figure7 shows the projection of beer samples onto the first two principal components. Most observations are distributed along PC1, reflecting differences in overall sensory quality, while PC2 captures variation in alcohol content. Although the observations form a continuous distribution with some overlap, the PCA projection provides a suitable low-dimensional representation for subsequent K-Means clustering.

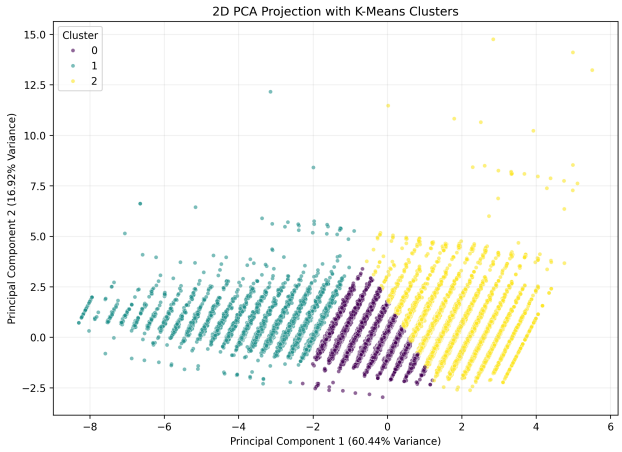


Figure 7: Beer Clusters Using PCA and K-Means

(a) PCA Variance Explained

Principal Component	Explained Variance	Cumulative Variance
PC1	0.6044	0.6044
PC2	0.1692	0.7736
PC3	0.1032	0.8768
PC4	0.0770	0.9539
PC5	0.0461	1.0000

(b) PCA Component Loadings

	PC1	PC2	PC3	PC4	PC5
review_aroma	0.852	-0.062	-0.140	0.463	-0.193
review_appearance	0.762	-0.119	0.634	-0.007	0.051
review_palate	0.852	-0.158	-0.157	-0.410	-0.238
review_taste	0.879	-0.152	-0.263	-0.020	0.366
beer_abv	0.467	0.883	0.002	-0.046	0.014

Figure 8: PCA Results: Variance breakdown and numerical feature loadings across the five principal components.

4.3.2 K-Means Clustering

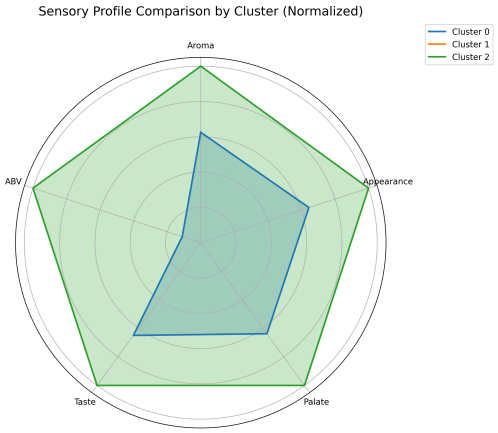
K-means clustering was applied to the standardized sensory variables to identify groups of beers with similar sensory characteristics. Although silhouette analysis indicated that k=2 produced the highest silhouette score, k=3 was selected as the final clustering solution because it generated more interpretable and practically meaningful beer profiles while maintaining acceptable cluster separation.

Table 9a summarizes the characteristics of the three clusters. Cluster 0 contained 23,235 beers and represented beers with moderate sensory ratings across all review attributes together with an average alcohol content of approximately **6.10%** ABV. Cluster 1, consisting of 6,803 beers, exhibited the lowest average ratings for aroma, appearance, palate, and taste, together with the lowest average alcohol content (5.80% ABV), representing comparatively lower-quality beers. In contrast, Cluster 2 contained 19,962 beers and achieved the highest average ratings across all sensory dimensions while also exhibiting the highest average alcohol content (8.52% ABV), suggesting a group of stronger, premium-quality beers.

The radar chart show in Figure 9b illustrates the normalised sensory profiles of the three clusters. Cluster 2 consistently recorded the highest values across aroma, appearance, palate, taste, and ABV, whereas Cluster 1 exhibited the weakest sensory profile across all attributes. Cluster 0 occupied an intermediate position between these two extremes. Overall, the clustering results indicate that beers naturally group according to overall sensory quality and alcohol strength rather than being differentiated by a single review characteristic.

(a) Cluster Profile Summary

Cluster	Size	review_aroma	review_appearance	review_palate	review_taste	beer_abv
0	23235	3.633000	3.774000	3.653000	3.701000	6.097000
1	6803	2.626000	2.973000	2.660000	2.548000	5.797000
2	19962	4.237000	4.216000	4.217000	4.323000	8.516000



(b) Sensory Profile Radar Chart

Figure 9: Cluster Profile Comparison: Mean feature values and normalized sensory visualization.

5 Conclusion

This study applied supervised and unsupervised machine learning methods to analyze the BeerAdvocate dataset through regression, classification, and clustering tasks. Overall, the results demonstrate that sensory review attritubes provide strong predictive information for overall beer ratings and reveal meaningful patters in beer characteristics.

For the Regression task, Linear Regression achieved the best performance, with the lowest RMSE and highest R^2, indicating that the relationship between the selected sensory features and overall rating is largely linear. For the Classification task, Logistic Regression produced the highest accuracy, outperforming Random Forest, LDA, and QDA in predicting beer rating tiers. In both tasks, review_taste and review_aroma were identified as the most influential predictors of beer ratings.

For the unsupervised learning task, PCA showed that the first principal component primarily represented overall sensory quality, while the second component reflected variation in alcohol content. K-Means clustering identified three interpretable groups corresponding to lower-quality beers, moderate-quality beers, and premium higher-quality beers, suggesting that beers naturally cluster according to both sensory quality and alcohol strength.

Several limitations should be acknowledged. The BeerAdvocate dataset contains self-selected online reviews, which may introduce sampling bias, and the target variable exhibited class imbalance, particularly for lower-rated beers. In addition, the strong relationship between the sensory review scores and the overall rating presents a potential data leakage concern if predictor selection is not carefully controlled. Finally, due to computational and time constraints, only a limited range of hyperparameters was explored for some models.

Despite these limitations, the results consistently demonstrate that sensory review attributes are highly effective for predicting beer quality and identifying meaningful groups of beers. The agreement between regression, classification, and clustering analyses provides confidence that taste and aroma are the primary factors influencing overall beer evaluations within the BeerAdvocate dataset.

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